Practical - 3

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Class : SYIT	Date/Time of Submission : 29-06-2026

a) Write a program to perform basic operations , indexing and slicing on arrays

Code:

```
import array
arr=array.array('i',[10,20,30,40,50])
print("Orginal Array:",arr)
arr.append(60)
print("After appending 60:",arr)
arr.insert(2,25)
print("After inserting 25 at index 2:",arr)
arr.remove(40)
print("After removing:",arr)
popped_element=arr.pop()
print("After popping last element(",popped_element,"):",arr)
print("element at index0:",arr[0])
print("element at index-1(last element):",arr[-1])
print("Slice from index 1 to 4:",arr[1:5])
print("Slice with step of2:",arr[::2])
print("reverse the arary using slicing:",arr[::-1])

print("Himanshu S037")
```

Output:

```
Orginal Array: array('i', [10, 20, 30, 40, 50])
After inserting 25 at index 2: array('i', [10, 20, 25, 30, 40, 50, 60])
After removing: array('i', [10, 20, 25, 30, 50, 60])
After popping last element( 60 ): array('i', [10, 20, 25, 30, 50])
element at index0: 10
element at index-1(last element): 50
Slice from index 1 to 4: array('i', [20, 25, 30, 50])
Slice with step of2: array('i', [10, 25, 50])
reverse the arary using slicing: array('i', [50, 30, 25, 20, 10])
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```

b) Write a program to implement mathematical function in array

Code:

```python
import numpy as np
arr=np.array([1,4,9,16,25])
print("Original Araay:",arr)
print("Square Root:",np.sqrt(arr))
print("Square:",np.square(arr))
print("Cube:",np.power(arr,3))
print("Log(base e):",np.log(arr))
print("Exponential(e^x):",np.exp(arr))
print("Sine:",np.sin(arr))
print("Cosine:",np.cos(arr))
print("Maximum value:",np.max(arr))
print("Minimum value:",np.min(arr))
print("Sum of elements:",np.sum(arr))
print("Mean(Average):",np.mean(arr))

print("Himanshu S037")
```

Output:

```
Original Araay: [ 1  4  9 16 25]
Square Root: [1. 2. 3. 4. 5.]
Square: [  1  16  81 256 625]
Cube: [    1    64   729  4096 15625]
Log(base e): [0.         1.38629436 2.19722458 2.77258872 3.21887582]
Exponential(e^x): [2.71828183e+00 5.45981500e+01 8.10308393e+03 8.88611052e+06
 7.20048993e+10]
Sine: [ 0.84147098 -0.7568025   0.41211849 -0.28790332 -0.13235175]
Cosine: [ 0.54030231 -0.65364362 -0.91113026 -0.95765948  0.99120281]
Maximum value: 25
Minimum value: 1
Sum of elements: 55
Mean(Average): 11.0
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```

c) Write a program to perform aliasing and copying

Code:

```python
import numpy as np

print("----- Array Aliasing using NumPy -----")
arr1 = np.array([11,22,33,44])
arr2 = arr1
print("Original arr1:", arr1)
print("Aliased arr2:", arr2)


arr2[1] = 99
print("After Modifying arr2[1] = 99")
print("arr1 (also changed):", arr1)
print("arr2:", arr2)


print("\n----- Array Copying Using NumPy -----")
arr3 = np.array([51,52,53,54])
arr4 = arr3.copy() # independent copy
print("Original arr3:", arr3)
print("Copied arr4:", arr4)


arr4[2] = 100
print("After modifying arr4[2] = 100")
print("arr3 (unchanged):", arr3)
print("arr4:", arr4)
print("Himanshu S037")
```

Output:

```
----- Array Aliasing using NumPy -----
Original arr1: [11 22 33 44]
Aliased arr2: [11 22 33 44]
After Modifying arr2[1] = 99
arr1 (also changed): [11 99 33 44]
arr2: [11 99 33 44]

----- Array Copying Using NumPy -----
Original arr3: [51 52 53 54]
Copied arr4: [51 52 53 54]
After modifying arr4[2] = 100
arr3 (unchanged): [51 52 53 54]
arr4: [ 51  52 100  54]
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```